ND2Tetris / Hardware Simulator / And.hdl

HDL --LBSt | Project 1 | And

```
1  // This file is part of www.nand2tetris.org
2  // and the book "The Elements of Computing Systems"
3  // by Nisan and Schocken, MIT Press.
4  // File name: projects/1/And.hdl
5  /**
6   * And gate:
7   * if (a and b) out = 1, else out = 0
8   */
9  CHIP And {
10     IN a, b;
11     OUT out;
12
13     PARTS:
14     Nand(a=a , b=b , out=nandOut );
15 }
```

Chip And | Eval | Reset | Clock: 0

Input pins
a 1
b 1

Output pins
out 0

Internal pins
nandOut 0

Test | Steps: 7 Outputs: 5 | Slow | Fast

Test Script | Compare File | Output File | Diff Table

```
21 set b 0,
22 eval,
23 output;
24
25 set a 1,
26 set b 1,
27 eval,
```

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Search

26°C

AND2Tetris / Hardware Simulator / Or.hdl

HDL --LBSt | Project 1 | Or

```
/**
 * Or gate:
 * if (a or b) out = 1, else out = 0
 */
CHIP Or {
    IN a, b;
    OUT out;

    PARTS:
    Nand(a=a, b=a, out=notA);
    Nand(a=b, b=b, out=notB);
    Nand(a=notA, b=notB, out=out);


}
```

Chip Or — Eval Reset Clock: 0

Input pins
- a 1
- b 1

Output pins
- out 1

Internal pins
- notA 0
- notB 0

Test — Steps: 7 Outputs: 5 Slow — Fast

Test Script | Compare File | Output File | Diff Table

```
set b 0,
eval,
output;

set a 1,
set b 1,
eval,
```

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HDL --LBSt | Project 1 | Xor

```
/**
 */
CHIP Xor {
    IN a, b;
    OUT out;

    PARTS:
    Nand (a=a, b=b, out=x);
    Nand(a=a , b=x , out=y );
    Nand(a=b , b=x , out=z );
    Nand(a=y , b=z , out=out );


}
```

Chip Xor | Eval Reset Cloc

Input pins
- a 0
- b 0

Output pins
- out 0

Internal pins
- x 1
- y 1
- z 1

Test | Steps: 7 Outputs: 0 Slow Fast

Test Script | Compare File | Output File | Diff Table

```
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/Xor.tst

load Xor.hdl,
compare-to Xor.cmp,
output-list a b out;
```

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How Google uses data.

NAND2Tetris / Hardware Simulator / Mux.hdl

HDL --LBSt | Project 1 | Mux

```
/**
 * Multiplexor:
 * if (sel = 0) out = a, else out = b
 */
CHIP Mux {
    IN a, b, sel;
    OUT out;

    PARTS:
    Not (in=sel, out=c);
    And(a=c , b=c , out=w1 );
    And(a=b , b=sel , out=w2 );
    Or(a=w1 , b=w2 , out=out );

}
```

Chip Mux | Eval | Reset | Clock: 0

Input pins
a 0
b 0
sel 0

Output pins
out 1

Internal pins
c 1
w1 1

Test | Steps: 7 Outputs: 0 | Slow | Fast

Test Script | Compare File | Output File | Diff Table

```
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/Mux.tst

load Mux.hdl,
compare-to Mux.cmp,
output-list a b sel out;
```

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NAND2Tetris / Hardware Simulator / DMux.hdl

HDL --LBSt | Project 1 | DMux

```
/**
 * [a, b] = [in, 0] if sel = 0
 *          [0, in] if sel = 1
 */
CHIP DMux {
    IN in, sel;
    OUT a, b;

    PARTS:
    Not(in=sel, out=notSel);
    And(a=in, b=notSel, out=a);
    And(a=in, b=sel, out=b);

    //// Replace this comment with your code.
}
```

Chip DMux | Eval Reset Clock: 0

Input pins
- in 0
- sel 0

Output pins
- a 0
- b 0

Internal pins
- notSel 1

Test | Steps: 7 Outputs: 0 Slow — Fast

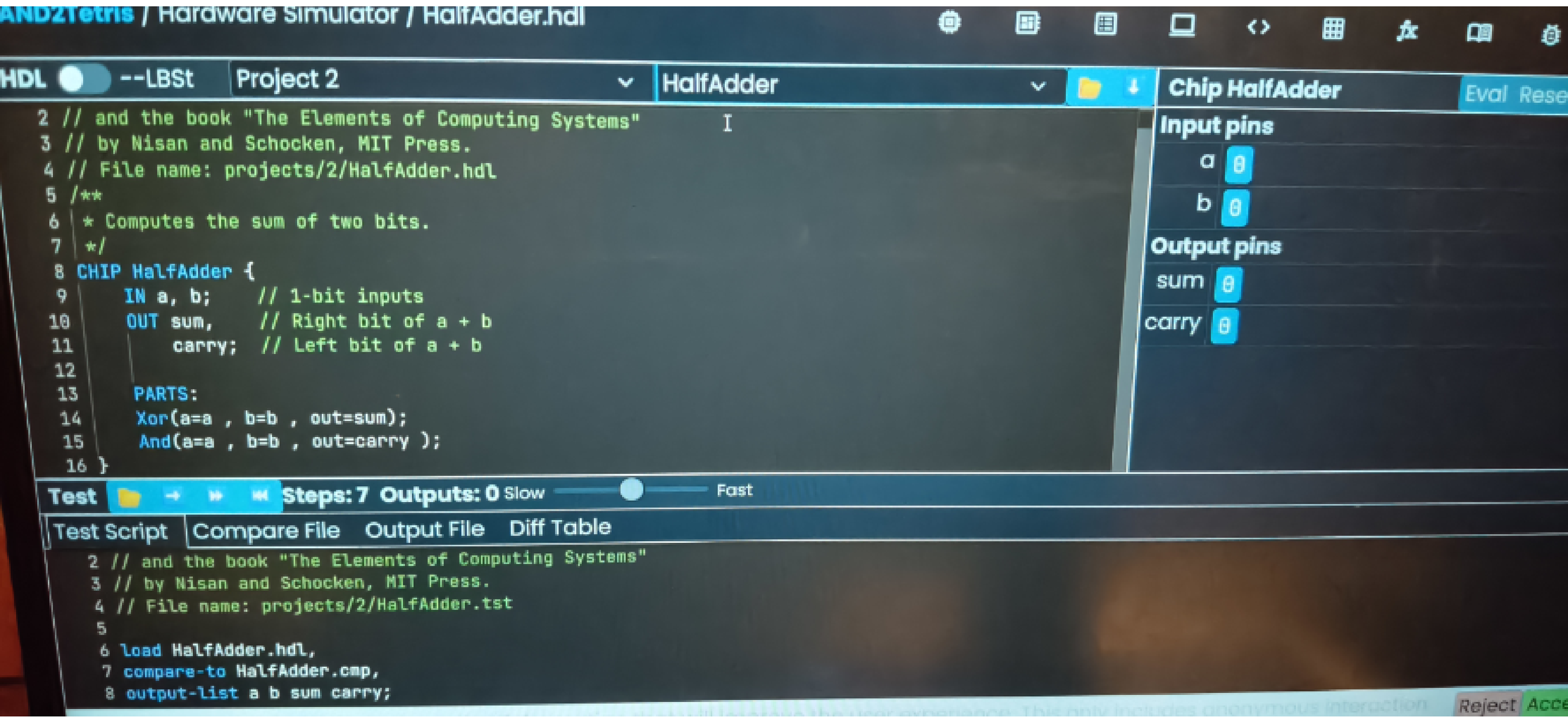
Test Script | Compare File | Output File | Diff Table

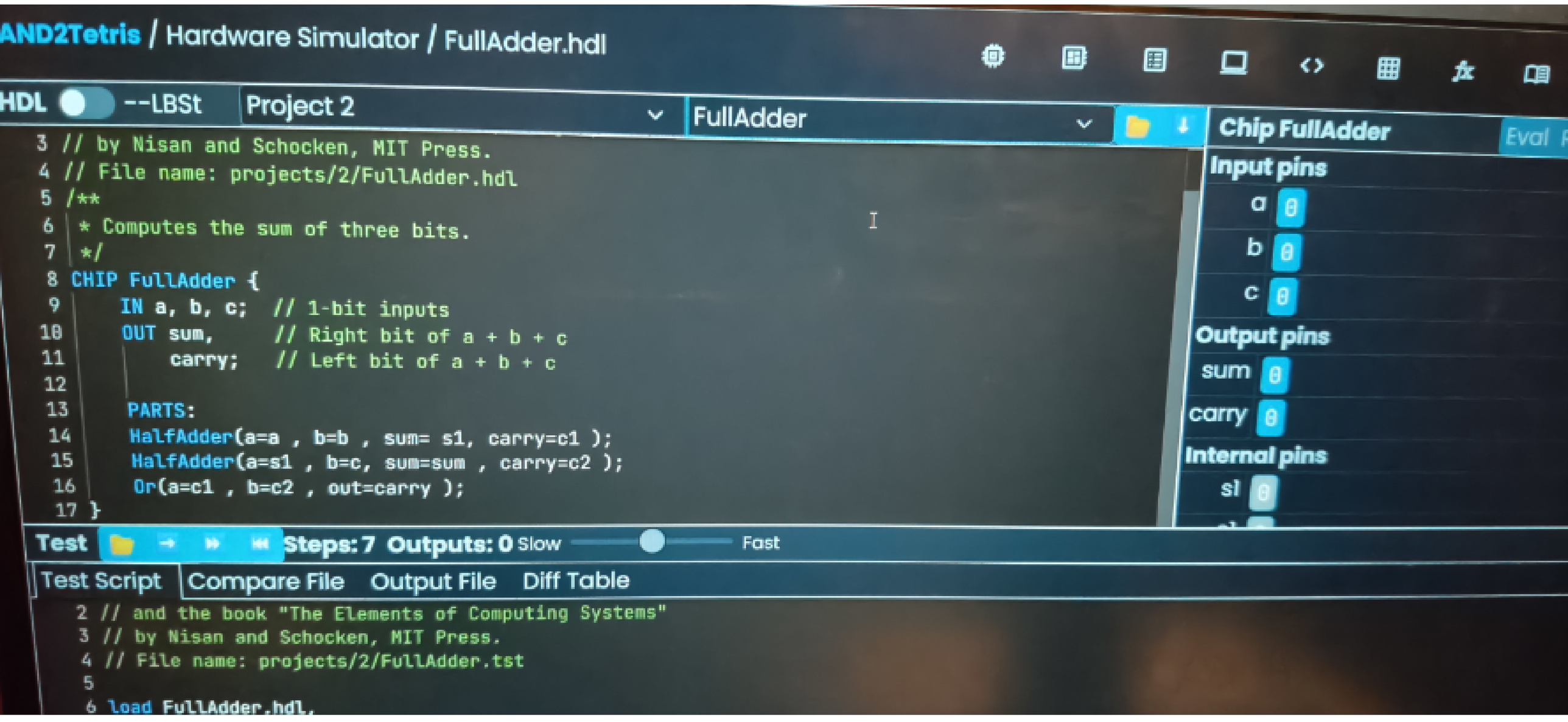
```
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/DMux.tst

load DMux.hdl,
compare-to DMux.cmp,
output-list in sel a b;
```

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Reject Accept X

26°C Search


AND2Tetris / Hardware Simulator / HalfAdder.hdl
HDL --LBSt
Project 2
HalfAdder
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/2/HalfAdder.hdl
/**
* Computes the sum of two bits.
*/
CHIP HalfAdder {
IN a, b; // 1-bit inputs
OUT sum, // Right bit of a + b
carry; // Left bit of a + b
PARTS:
Xor(a=a , b=b , out=sum);
And(a=a , b=b , out=carry);
}
Chip HalfAdder
Eval
Input pins
a 0
b 0
Output pins
sum 0
carry 0
Test
Steps: 7 Outputs: 0
Slow
Fast
Test Script
Compare File
Output File
Diff Table
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/2/HalfAdder.tst
load HalfAdder.hdl,
compare-to HalfAdder.cmp,
output-list a b sum carry;
Reject


AND2Tetris / Hardware Simulator / FullAdder.hdl
HDL --LBSt
Project 2
FullAdder
// by Nisan and Schocken, MIT Press.
// File name: projects/2/FullAdder.hdl
/**
* Computes the sum of three bits.
*/
CHIP FullAdder {
IN a, b, c; // 1-bit inputs
OUT sum, // Right bit of a + b + c
carry; // Left bit of a + b + c
PARTS:
HalfAdder(a=a , b=b , sum= s1, carry=c1);
HalfAdder(a=s1 , b=c, sum=sum , carry=c2);
Or(a=c1 , b=c2 , out=carry);
}
Chip FullAdder
Input pins
a 0
b 0
c 0
Output pins
sum 0
carry 0
Internal pins
s1 0
Test
Steps: 7 Outputs: 0 Slow Fast
Test Script
Compare File
Output File
Diff Table
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/2/FullAdder.tst
load FullAdder.hdl,

--LBSt | Project 2 | Add16

```
CHIP Add16 {
    FullAdder(a=a[5], b=b[5], c=c5, sum=out[5], carry=c6);
    FullAdder(a=a[6], b=b[6], c=c6, sum=out[6], carry=c7);
    FullAdder(a=a[7], b=b[7], c=c7, sum=out[7], carry=c8);
    FullAdder(a=a[8], b=b[8], c=c8, sum=out[8], carry=c9);
    FullAdder(a=a[9], b=b[9], c=c9, sum=out[9], carry=c10);
    FullAdder(a=a[10], b=b[10], c=c10, sum=out[10], carry=c11);
    FullAdder(a=a[11], b=b[11], c=c11, sum=out[11], carry=c12);
    FullAdder(a=a[12], b=b[12], c=c12, sum=out[12], carry=c13);
    FullAdder(a=a[13], b=b[13], c=c13, sum=out[13], carry=c14);
    FullAdder(a=a[14], b=b[14], c=c14, sum=out[14], carry=c15);
    FullAdder(a=a[15], b=b[15], c=c15, sum=out[15], carry=overflow);

    //// Replace this comment with your code.
}
```

Chip Add16

Input pins

a 0 0 0 0 0

b 0 0 0 0 0

Output pins

out 0 0 0 0 0

Internal pins

c1 0

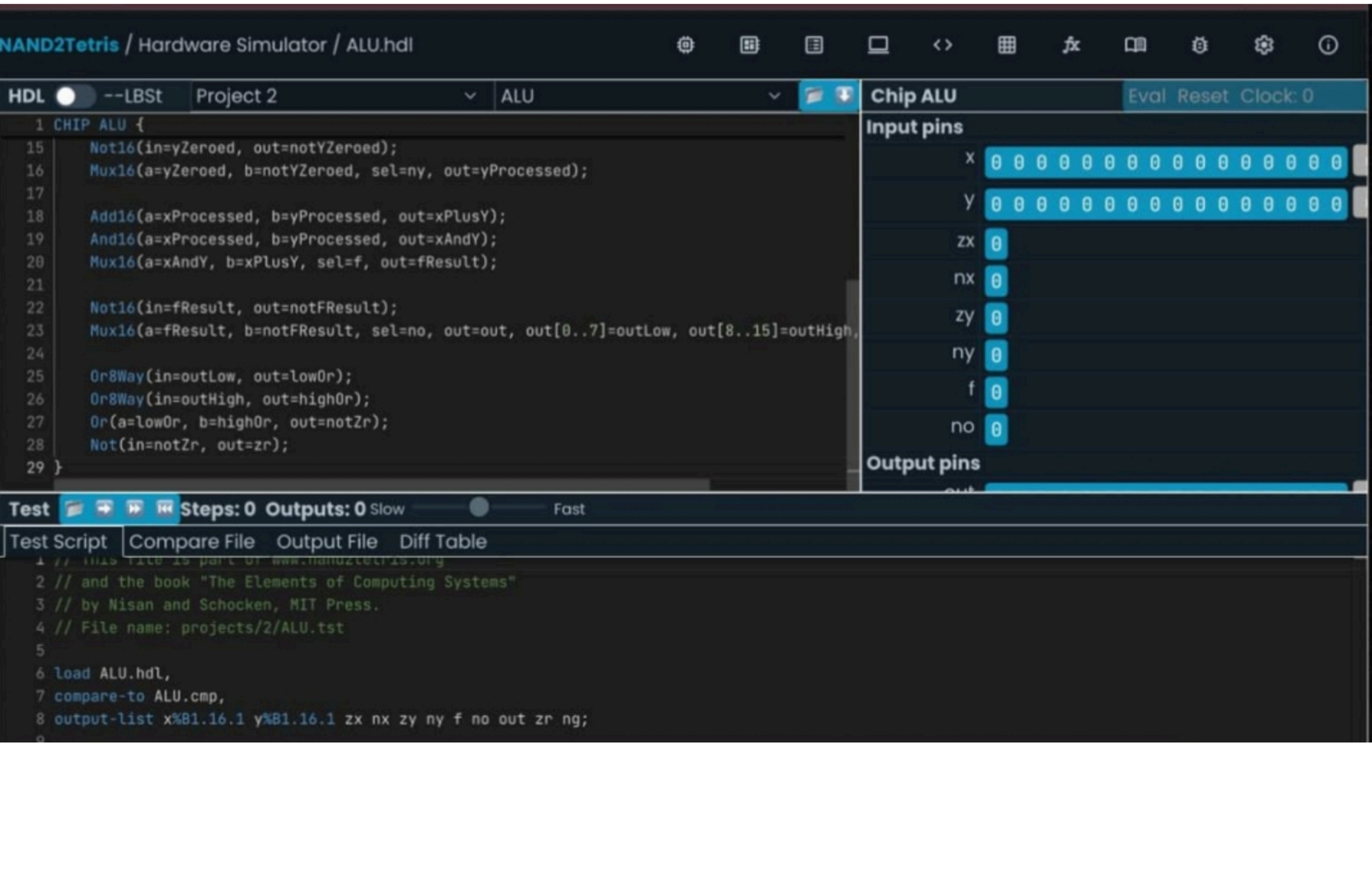
c2 0

c3 0

Test | Steps: 7 Outputs: 0 Slow Fast

Test Script | Compare File | Output File | Diff Table

```
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/2/Add16.tst
```

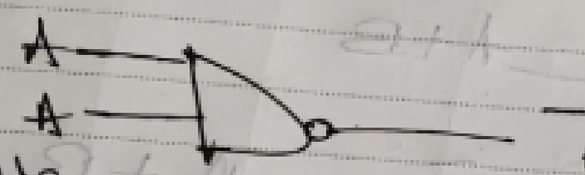

NAND2Tetris / Hardware Simulator / ALU.hdl
HDL --LBSt
Project 2
ALU
1 CHIP ALU {
15 Not16(in=yZeroed, out=notYZeroed);
16 Mux16(a=yZeroed, b=notYZeroed, sel=ny, out=yProcessed);
17
18 Add16(a=xProcessed, b=yProcessed, out=xPlusY);
19 And16(a=xProcessed, b=yProcessed, out=xAndY);
20 Mux16(a=xAndY, b=xPlusY, sel=f, out=fResult);
21
22 Not16(in=fResult, out=notFResult);
23 Mux16(a=fResult, b=notFResult, sel=no, out=out, out[0..7]=outLow, out[8..15]=outHigh,
24
25 Or8Way(in=outLow, out=lowOr);
26 Or8Way(in=outHigh, out=highOr);
27 Or(a=lowOr, b=highOr, out=notZr);
28 Not(in=notZr, out=zr);
29 }
Chip ALU
Eval Reset Clock: 0
Input pins
x 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
y 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
zx 0
nx 0
zy 0
ny 0
f 0
no 0
Output pins
Test
Steps: 0 Outputs: 0 Slow
Fast
Test Script
Compare File
Output File
Diff Table
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/2/ALU.tst
6 load ALU.hdl,
7 compare-to ALU.cmp,
8 output-list x%B1.16.1 y%B1.16.1 zx nx zy ny f no out zr ng;

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Date: / /

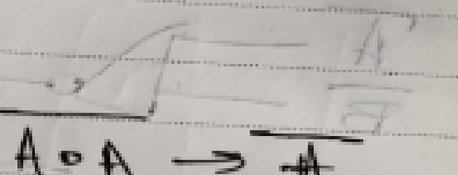
1.) NAND to NOT (do not input same logic)

•1 Input, 1 NAND



Truth table

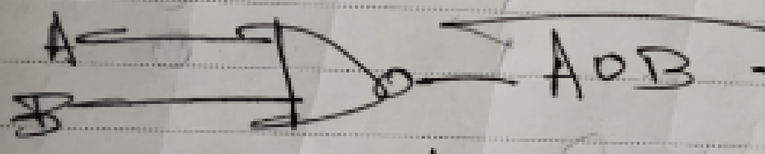
A	NAND A
0	1
1	0



A·A NAND

A ⊼ A = A

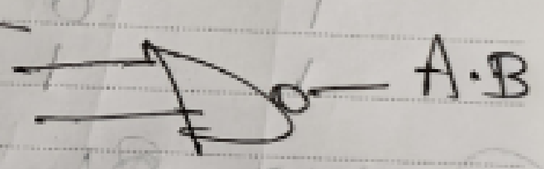
2.) NAND & to AND



2 Input
2 NAND

Truth table

A	B	AND	NOT(AND)-WAND
0	0	0	1
0	1	0	1
1	0	0	1
1	1	1	0

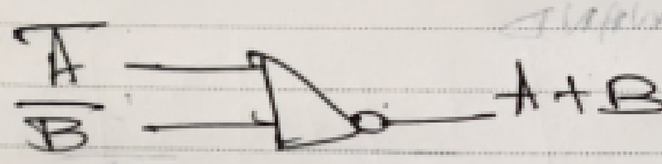


Su Mo Tu We Th Fr Sa

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Date: / /

NAND & to OR



$\overline{A \cdot B} \rightarrow \overline{A} + \overline{B} = A + B$

Truth Table:

A	B	OR	NOR
0	0	0	1
0	1	1	0
1	0	1	0
1	1	1	0

(4) NAND & to XOR

XOR have different ~~output~~ input → output = 1

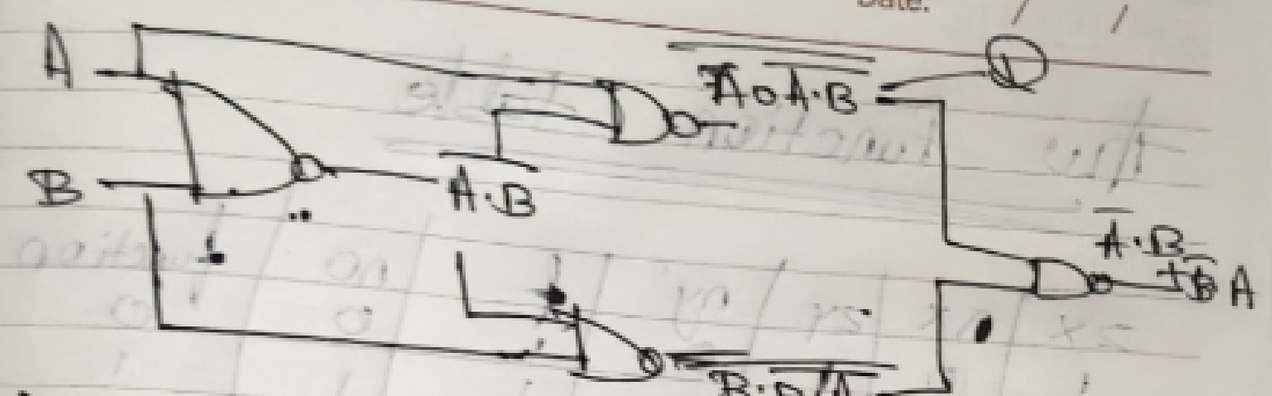
Same input → output = 0

Truth table

A	B	output
0	0	0
0	1	1
1	0	1
1	1	0

Su Mo Tu We Th Fr Sa

Date: / /



for eq ①

$\overline{(A \cdot \overline{(A \cdot B}}$

$A \cdot \overline{A} + A \cdot \overline{B}$

$\overline{A \cdot \overline{B}}$

$= \overline{A} + \overline{\overline{B}} = \overline{A} + B$

$(\overline{A} + B) \cdot (\overline{B} + A)$

$\overline{A} \cdot \overline{B} + \overline{A} \cdot A + \overline{B} \cdot B + B \cdot A$

$\overline{A \cdot B} + B \cdot A$

$\overline{A} + \overline{B} + \overline{B} + \overline{A}$

$(A + B) + (\overline{B} + \overline{A})$

$A \cdot \overline{B} + A \cdot \overline{A} + B + \overline{B} + BA$

$A \cdot \overline{B} + B \cdot \overline{A}$

for ② B: $\overline{(B \cdot A)}$ ②

$B \cdot \overline{B} + B \cdot \overline{A}$

$B \cdot \overline{A}$

$\overline{B} + \overline{\overline{A}} = \overline{B} + A$.

We can't help everyone, but everyone can help someone. – Ronald Reagan

Su Mo Tu We Th Fr Sa

Date: / /

Alu function table

function	no	f	zy	zx	ax	zx
0	0	1	1	1	0	1
1	1	1	1	1	1	1
-1	2	1	1	1	0	1
x	3	1	0	1	1	1
y	4	1	1	0	0	1
!x	5	1	0	1	1	1
!y	6	1	1	0	0	1
x+y	7	0	0	0	0	0
x-y	8	1	0	0	1	0
y-x	9	1	0	0	1	0
x&y	10	0	0	0	0	0
x|y	11	1	0	0	1	0
1	12	1	1	1	0	1
-1	13	1	1	1	1	1
!(x+y)	14	1	0	0	1	0
x=y	15	0	0	0	0	0

Chance is a word void of sense; nothing can exist without a cause. – Voltaire